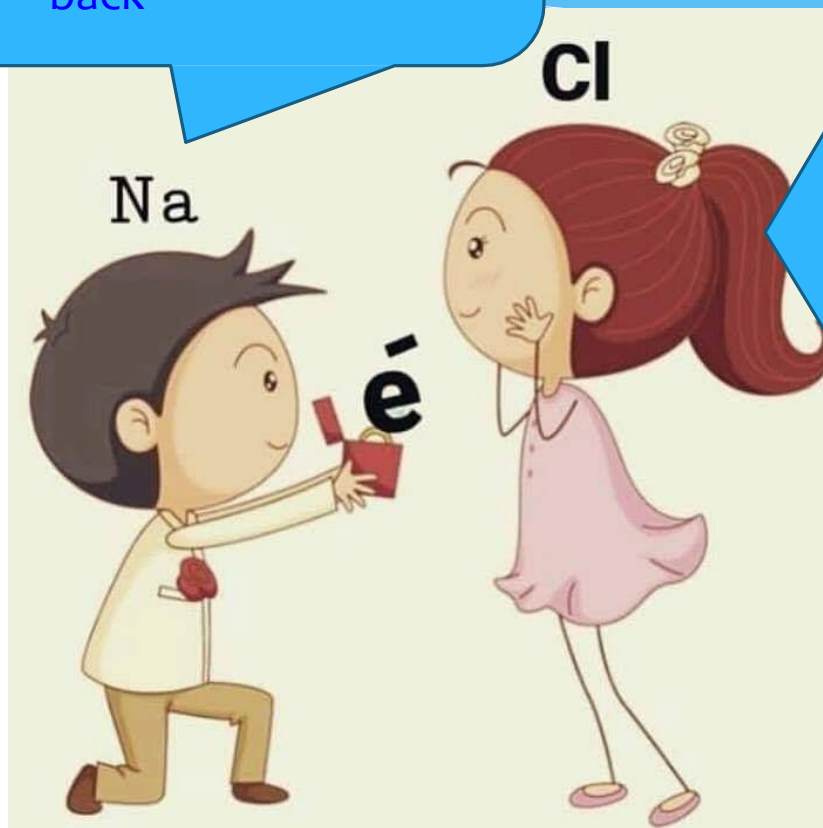




Bonding and Periodic Trends

By
Dr Bosire

Wow, Chlorine, you look mwaah, fantastic... I can give you my heart. But first, receive this gift from me, a beautiful electron. I will never claim it back



Oh dear Sodium, I love that. This ring of an electron will make our relationship reinforced... and forever attracted to each other!!

That is nice. Thank you! Now you will be known as Sodium Chloride!



Bonding



<https://www.youtube.com/watch?v=pjiD0FrUNN8>

Chemical bonding is an **energetically beneficial interaction** between atoms that requires them to maintain a specific distance from one another in space

Chemical bonds are the glue that **hold molecules together**

Atoms are so small, and when bonded the distances between them are so short, that a special unit of distance, called an angstrom, $\text{\AA}$, is used to measure bond lengths. One angstrom is one ten billionth of a meter.

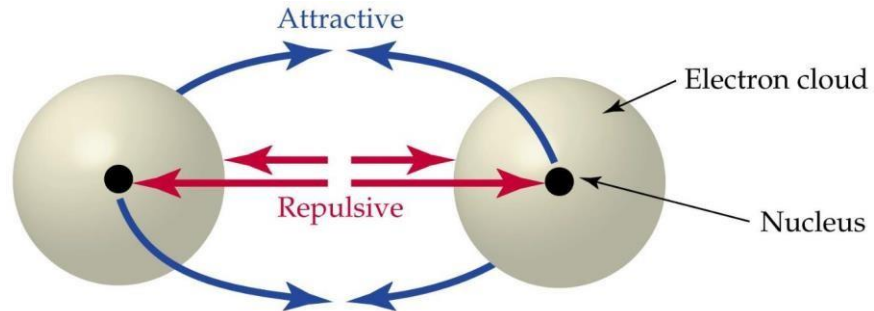
$$\text{\AA} = 10^{-10} \text{m}$$



A **bond** is a linkage between a particular pair of atoms e.g. H and Cl in HCl, N and H in NH_3 etc.



A chemical bond is the net result of attractive and repulsive electrostatic forces. When bringing together two atoms that are initially very far apart, three types of interactions occur, one attractive and two repulsive forces.



The nucleus-electron attractions are greater than the nucleus-nucleus and electron-electron repulsions, resulting in a net attractive force that holds the atoms together in a molecule.

A bond may be strong, weak, long, short, polar or non polar, single, double, triple and even quadruple.



One general rule governs the bonding process:

Through bonding, atoms attain a complete outer shell of valence electrons.

Alternatively, because the noble gases in column 8A of the periodic table are especially stable as atoms having a filled shell of valence electrons, the general rule can be restated as:

Through bonding, atoms attain a stable noble gas configuration of electrons.

1A																	8A				
1 H 1s ¹	2A															2 He 1s ²					
3 Li 2s ¹	4 Be 2s ²															3A 5 B 2s ² 2p ¹	4A 6 C 2s ² 2p ²	5A 7 N 2s ² 2p ³	6A 8 O 2s ² 2p ⁴	7A 9 F 2s ² 2p ⁵	10 Ne 2s ² 2p ⁶
11 Na 3s ¹	12 Mg 3s ²	3B	4B	5B	6B	7B	8B			1B	2B	13 Al 3s ² 3p ¹	14 Si 3s ² 3p ²	15 P 3s ² 3p ³	16 S 3s ² 3p ⁴	17 Cl 3s ² 3p ⁵	18 Ar 3s ² 3p ⁶				
19 K 4s ¹	20 Ca 4s ²	21 Sc 3d ¹ 4s ²	22 Ti 3d ² 4s ²	23 V 3d ³ 4s ²	24 Cr 3d ⁵ 4s ¹	25 Mn 3d ⁵ 4s ²	26 Fe 3d ⁶ 4s ²	27 Co 3d ⁷ 4s ²	28 Ni 3d ⁸ 4s ²	29 Cu 3d ¹⁰ 4s ¹	30 Zn 3d ¹⁰ 4s ²	31 Ga 4s ² 4p ¹	32 Ge 4s ² 4p ²	33 As 4s ² 4p ³	34 Se 4s ² 4p ⁴	35 Br 4s ² 4p ⁵	36 Kr 4s ² 4p ⁶				
37 Rb 5s ¹	38 Sr 5s ²	39 Y 4d ¹ 5s ²	40 Zr 4d ² 5s ²	41 Nb 4d ⁴ 5s ¹	42 Mo 4d ⁵ 5s ¹	43 Tc 4d ⁵ 5s ²	44 Ru 4d ⁷ 5s ¹	45 Rh 4d ⁸ 5s ¹	46 Pd 4d ¹⁰	47 Ag 4d ¹⁰ 5s ¹	48 Cd 4d ¹⁰ 5s ²	49 In 5s ² 5p ¹	50 Sn 5s ² 5p ²	51 Sb 5s ² 5p ³	52 Te 5s ² 5p ⁴	53 I 5s ² 5p ⁵	54 Xe 5s ² 5p ⁶				
55 Cs 6s ¹	56 Ba 6s ²	57 *La 5d ¹ 6s ²	72 Hf 5d ² 6s ²	73 Ta 5d ³ 6s ²	74 W 5d ⁴ 6s ²	75 Re 5d ⁵ 6s ²	76 Os 5d ⁶ 6s ²	77 Ir 5d ⁷ 6s ²	78 Pt 5d ⁹ 6s ¹	79 Au 5d ¹⁰ 6s ¹	80 Hg 5d ¹⁰ 6s ²	81 Tl 6s ² 6p ¹	82 Pb 6s ² 6p ²	83 Bi 6s ² 6p ³	84 Po 6s ² 6p ⁴	85 At 6s ² 6p ⁵	86 Rn 6s ² 6p ⁶				

A first-row element hydrogen can accommodate two electrons around it. This would make it like the noble gas helium at the end of the same row.

Second-row element, such as C, N, O and F are most stable with eight valence electrons around them like neon.

Elements that behave in this manner are said to follow the octet rule.



Two atoms are walking down the street. One atom says to the other atom, "I think I have lost an electron!" The other atom responds, "Are you sure?" The first atom replies, "I'm positive."



Hehe...interesting
Ionic Bonds

There are three major types of bonds

- ✓ Ionic Bonds,
- ✓ Covalent Bonds (including dative or coordinate bond)
- ✓ Metallic bonds.

NB: Atoms form molecules as away of attaining **chemical stability**. A molecule (combination of two or more atoms) only forms if it is chemically more stable and has a lower energy than individual atoms.

The most stable electronic arrangement is a **noble gas electronic structure** and most molecules have this arrangement.



This octet rule is broken in a significant number of cases:



In atoms such as **Be or B** which have less than four outer electrons. Even if all the outer electrons are used to form bonds an octet cannot be attained.

Where atoms have an extra energy level which is close in energy to the **p level**, which may accept electrons and be used in bonding. PF_3 obey the octet rule but PF_5 does not because the phosphorous uses one **3s**, **three 3p and one 3d** orbitals. These violations become increasingly common in elements after the first two periods of eight elements in the periodic table.

The octet rule does not work in molecules with an odd number of electrons such as **NO and ClO_2** .

Bonding theories



- ☐ *Lewis Theory*
- ☐ *Valence Shell Electron Pair Repulsion theory (VSEPR)*
- ☐ *Valence bond theory (VBT) and hybridization of atomic orbitals*
- ☐ *Molecular Orbital (MO) theory.*

Lewis Theory

Fundamental ideas in Lewis' theory:

Electrons, especially those of the outermost (valence) electronic shell, play a fundamental role in chemical bonding.

In some cases electrons are *transferred* from one atom to another. Positive and negative ions are formed and attract each other through electrostatic forces called **ionic bonds** or one or more pairs of electrons are shared between atoms; this *sharing* of electrons is called a **covalent bond**.

Electrons are *transferred, or shared, in such a way that each atom acquires an especially stable electron configuration*. Usually this is a noble gas configuration, one with eight outer shell electrons, or an **octet**.



Lewis Symbols and Lewis Structures



A **Lewis symbol** consists of a chemical symbol to represent the nucleus and *core* (inner shell) electrons of an atom, together with dots placed around the symbol to represent the *valence* (outer shell) electrons.

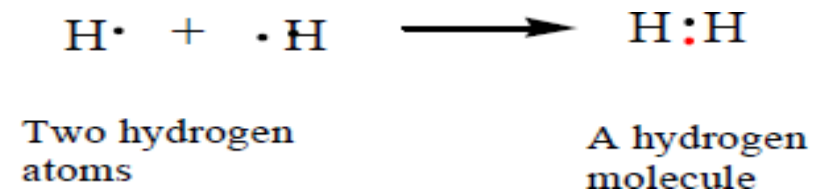
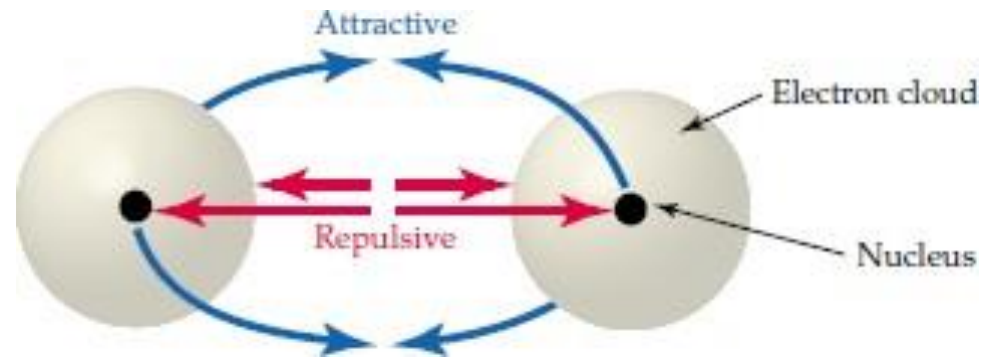
Lewis structure for: Na electron configuration $[\text{Ne}]3s^1$ is $\cdot\text{Na}$

O electron configuration $[\text{He}]2s^22p^4$ is $\cdot\ddot{\text{O}}\cdot$

A **Lewis structure** is a combination of Lewis symbols that represent either the transfer or sharing of electrons in a chemical bond.

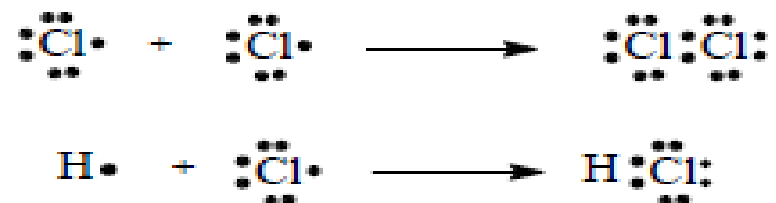


Covalent bonding (sharing of electrons) as in hydrogen molecule is written showing a pair of dots between hydrogen atoms, indicating that the hydrogen atom share the pair of electrons in a **covalent bond**. The combining can be represented as; follows:



The shared pair is usually represented by a line e.g. $\text{H}—\text{H}$

Atoms other than hydrogen also form covalent bonds. For example,

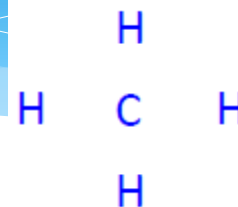




Draw a Lewis structure for methane, CH₄

Step 1: Arrange the atoms

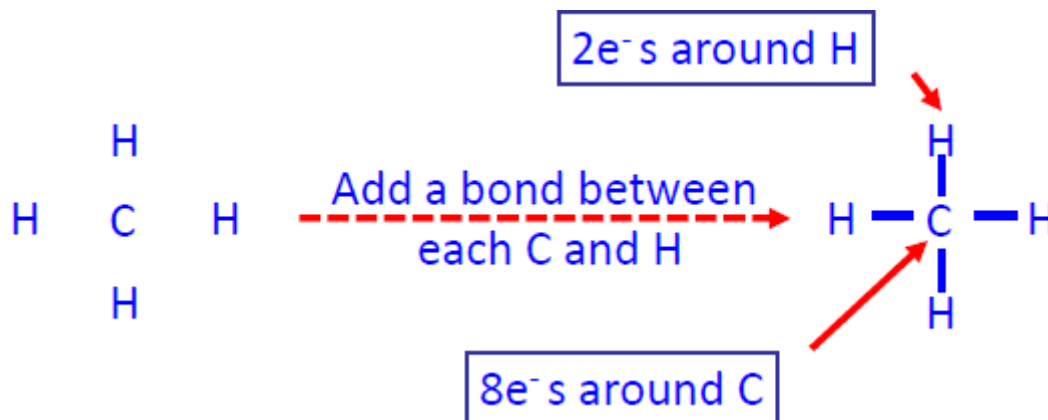
Place C in the center and 4 H's on the outside.



Step 2: Count the total number of valence electrons

$$\begin{aligned} 1\text{C} \times 4e^- &= 4e^-; 4\text{H} \times 1e^- = 4e^- \\ \text{Total} &= 8e^- \end{aligned}$$

Step 3: Add the bonds and lone pairs electrons.



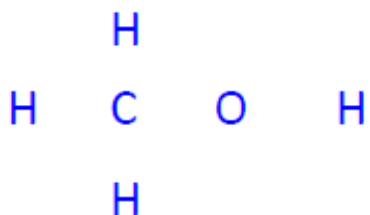
Adding four two electron bonds around carbon uses all eight valence electrons, and so there are no lone pairs



Draw a Lewis structure for methanol, a compound with molecular formula CH_4O



Step 1: Arrange the atoms

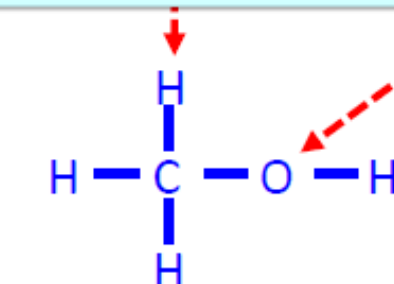
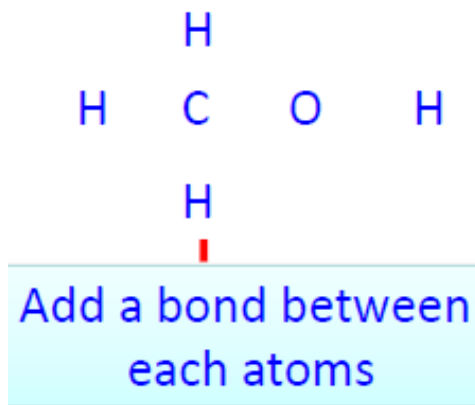


- H on the outside
- Four atoms around C
- Two atoms around O

Step 2: Count the total number of valence electrons

$$\begin{aligned} 1\text{C} \times 4e^- &= 4e^- \\ 1\text{O} \times 6e^- &= 6e^- \\ 4\text{H} \times 1e^- &= 4e^- \\ \hline \text{Total} &= 14e^- \end{aligned}$$

Step 3: Add the bonds and lone pairs electrons.

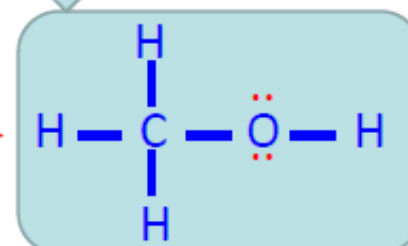


Only 10 e^- s used

This uses all 14 electrons, giving every H two electrons and every second-row element eight, C and O in this case.

No octet

Add lone pairs on O



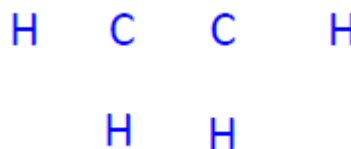
Valid structure



Draw a Lewis structure for C₂H₄



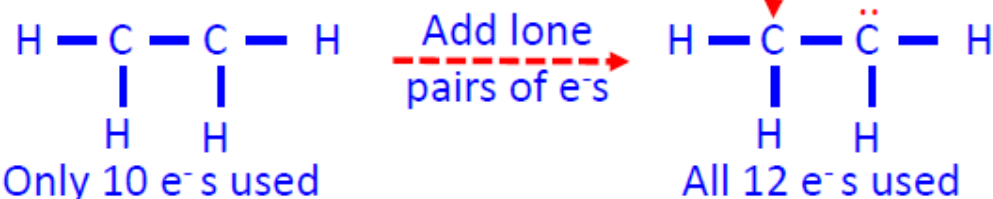
Step 1: Arrange the atoms



Step 2: Count the total number of valence electrons

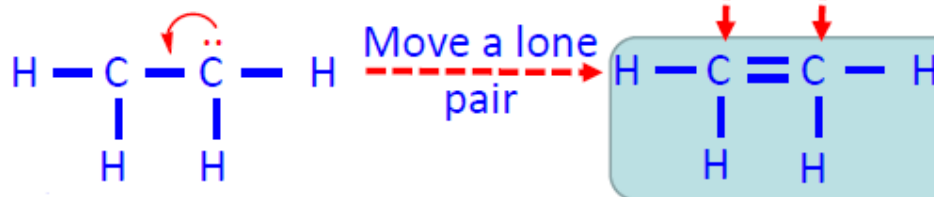
$$\begin{array}{r} 2\text{C} \times 4e^- = 8e^- \\ 4\text{H} \times 1e^- = 4e^- \\ \hline \text{Total} = 12e^- \end{array}$$

Step 3: Add the bonds and lone pairs electrons.

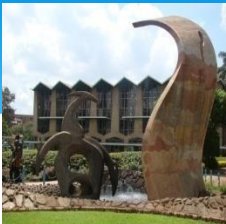


No octet

To give both C's an octet, change one lone pair into one bonding pair of electrons between the two C's, forming a double bond.



Each C has 4 bonds



Assignment: Draw a Lewis structure for C_2H_2



After placing all electrons in bonds and lone pairs, use a lone pair to form a multiple bond if an atom does not have an octet

The examples discussed above are for **neutral compounds**

However, compounds can also be either positively or negatively charged and we should be able to draw the structures of such compounds and assign charges to each individual atom in the particular compound

The decision as to where to put the charge is made by calculating the **formal charge** for each atom in an ion or a molecule

Formal charge is the charge assigned to individual atoms in a Lewis structure

$$\text{Formal charge} = \# \text{ of valence } e^{-} \text{ s} - \left(\# \text{ of nonbonding } e^{-} \text{ s} + \frac{1}{2} \# \text{ of bonding } e^{-} \text{ s} \right)$$



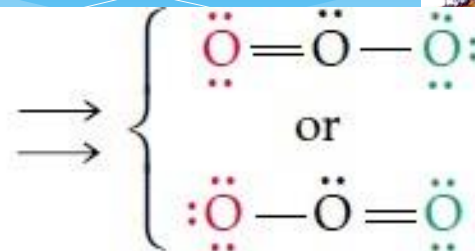
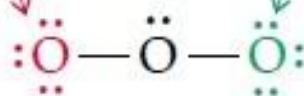
Lewis Structures and Resonance

The Lewis structure of ozone O_3 are:

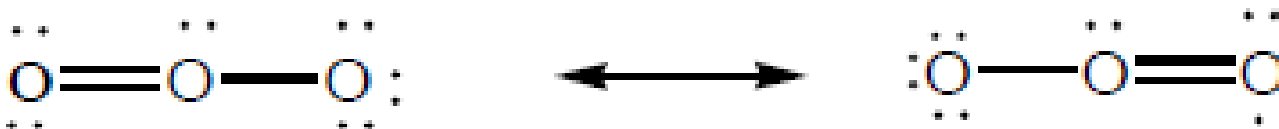


Move a lone pair from
this oxygen?

Or from this
oxygen?

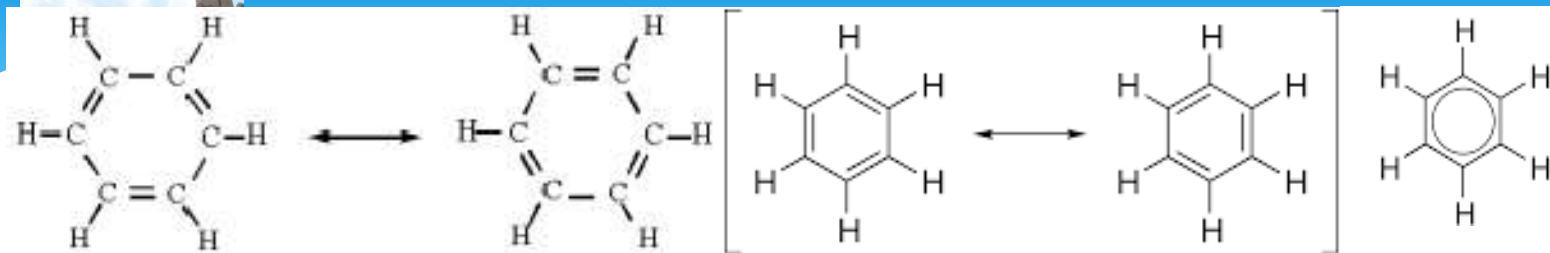


In fact, neither is correct by itself. Whenever it is possible to write more than one Lewis structure for a molecule the actual electronic structure is an average of the various possibilities called a **resonance hybrid**. Ozone does not have one $O=O$ double bond and one $O-O$ single bond as individual structures imply, rather ozone has two equivalent $O-O$ bonds that one can think of as having a *bond order* of 1.5 - midway between pure single bond and pure double bond. Both have an identical length of 1.28 Å.

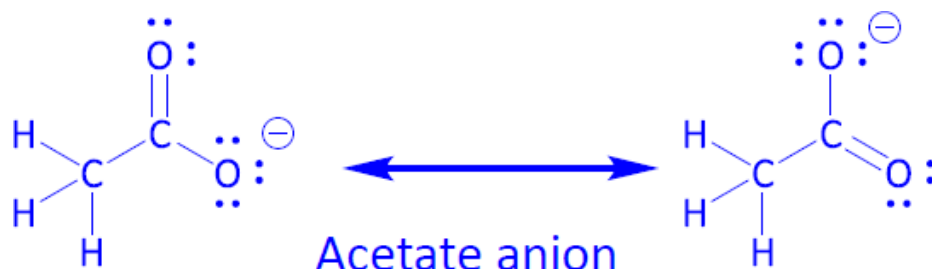
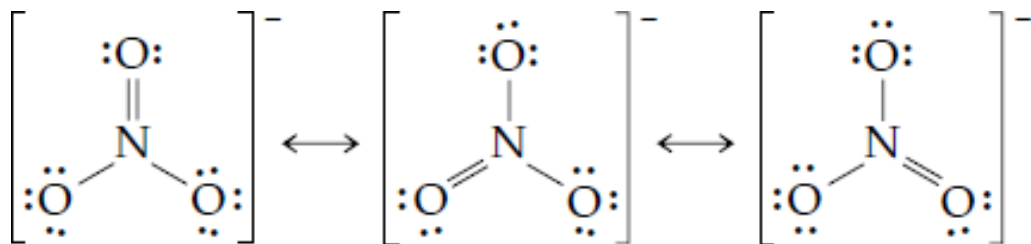


Class exercise: Write the resonance structures of benzene, acetate ion, carbonate ion and nitrate ion

Benzene resonance structures



Nitrate ion resonance structures



Draw the resonance structure of carbonate ion, CO_3^{2-} .

Note: Lewis structures say nothing about bond angles or shapes of molecules in general.



Valence Bond Theory (VBT)

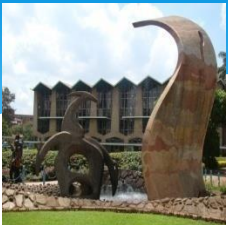


This was developed to use the methods of **quantum mechanics** to explain chemical bonding. It focuses on how the atomic orbitals of the individual atoms combine on molecular formation to give individual chemical bonds. It uses three principles to explain bonding:

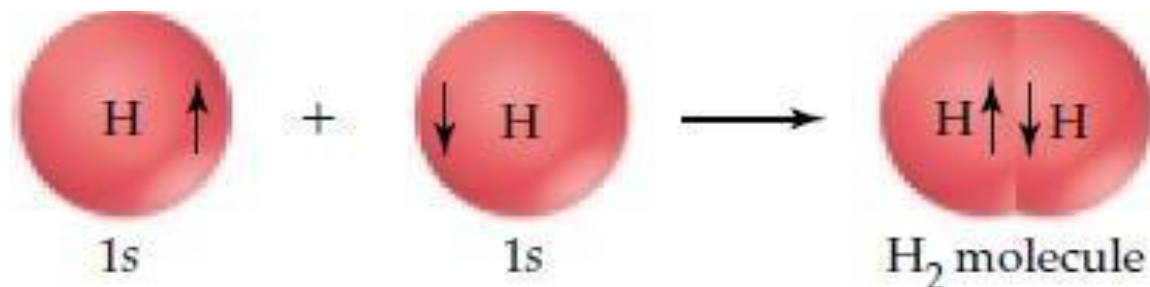
- a) **Overlap of atomic orbitals and**
- b) **Hybridization of atomic orbitals**
- c) **Resonance of molecular structures**

- ✓ Covalent bond occurs when atoms share electrons.
- ✓ The build up of electron density between two nuclei occurs when a valence atomic orbital of one atom overlaps with one of another.
- ✓ The orbitals share a region of space i.e. they overlap.
- ✓ The overlap of orbitals allows two electrons of opposite spin to share the common space between the nuclei forming a covalent bond.



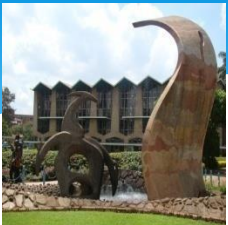


In the hydrogen molecule for instance the H-H bond results from overlap of two singly occupied H 1s orbitals. In the valence bond model, the strength of covalent bond depends on amount of overlap. **The greater the overlap, the stronger the bond.**

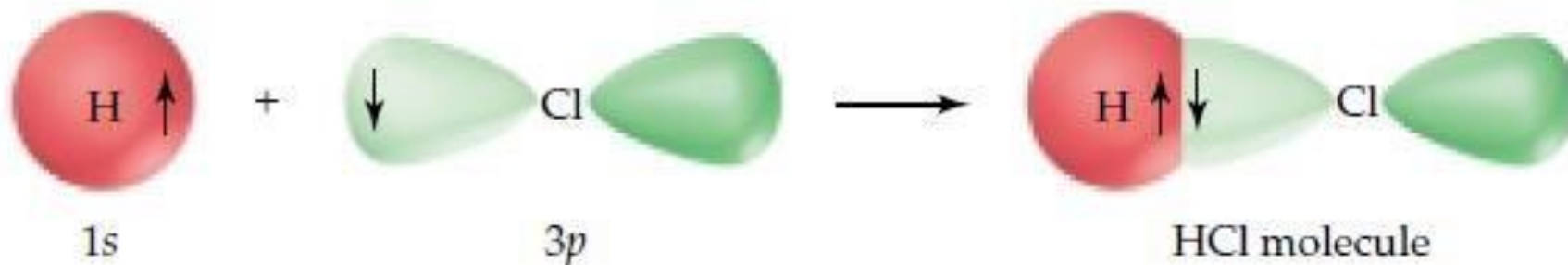


In the fluorine molecule, F_2 , for instance each atom has electron configuration: $[He]2s^2 2p_z^2 2p_y^2 2p_x^1$. This means that the fluorine bond results from overlap of two singly occupied two 2p orbitals. The 2p orbitals must point directly to one another for optimum overlap to occur, and the F-F form bond along the orbitals. Such bonds that result from head-on orbital overlap are called **sigma (σ) bonds**.



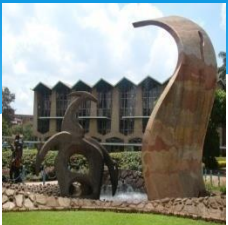


In HCl, the covalent bond involves overlap of 1s orbital (non-directional) with chlorine 3p orbital and formed along p-axis.



The key ideas of valence bond theory can be summarized by few statements:

- Covalent bonds are formed by **overlap of atomic orbitals**, each of which contains **1 electron of opposite spin**.
- Each of the bonded atoms maintains its own atomic orbitals, but the electron pair in the overlapping pair is shared by both atoms.
- The greater the amount of orbital overlap the stronger the bond. This leads to a directional character to the bond when other than s-orbitals are formed.



Molecular Geometry and Hybrid Orbitals



Hybridization (Hybrid Orbitals)

For polyatomic molecules we would like to be able to explain the number of bonds formed and the geometries of the molecules.

The number of bonds formed by an atom is usually the same as the number of unpaired electrons in the ground state. However in some cases the atom may form more bonds than this.

This occurs by excitation of the atom when electrons which were paired in the ground state are unpaired and promoted into suitable empty orbitals.

This increases the number of unpaired electrons available for bonding.

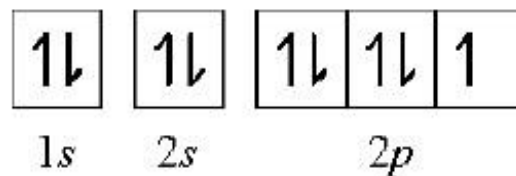


sp Hybrid Orbitals

Consider the Lewis structure of BeF₂:

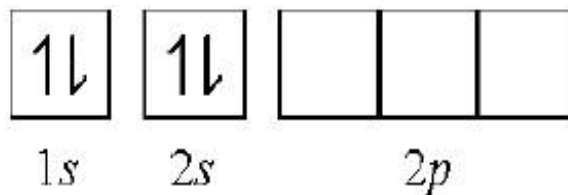


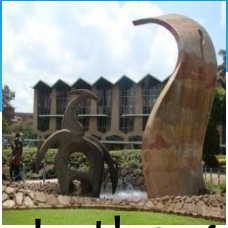
The fluorine atom electron configuration is $1s^2 2s^2 2p^5$



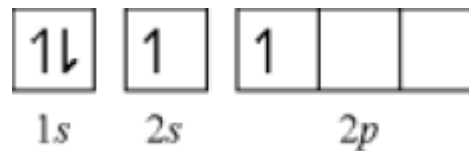
- There is an unpaired electron in a $2p$ orbital
- This unpaired $2p$ electron can be paired with an unpaired electron in the Be atom to form a covalent bond

The Be atom electron configuration is $1s^2 2s^2$





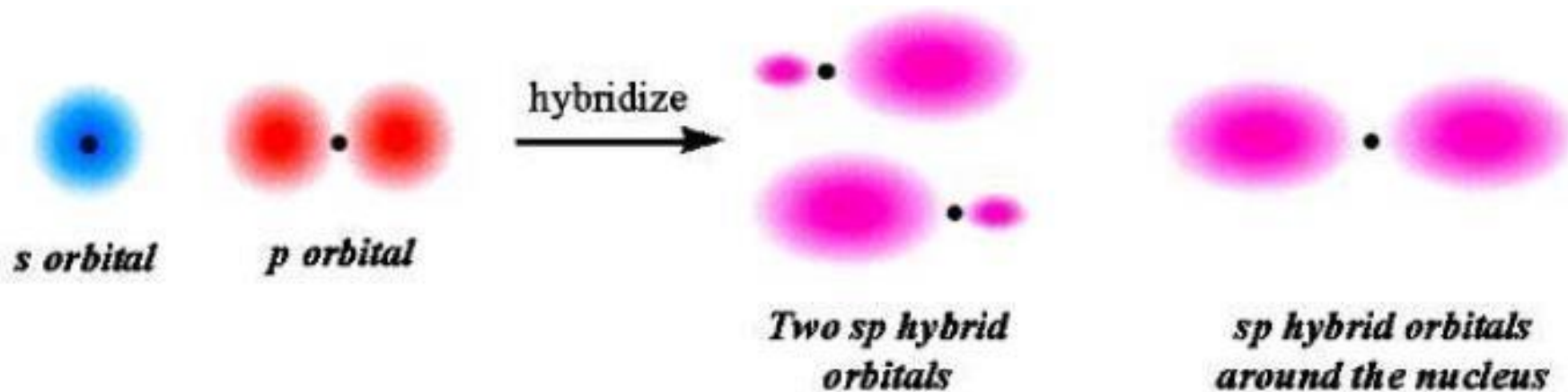
In the ground state, there are no unpaired electrons (the Be atom is incapable of forming a covalent bond with a fluorine atom. However, the Be atom could obtain an unpaired electron by promoting an electron from the 2s orbital to the 2p orbital:



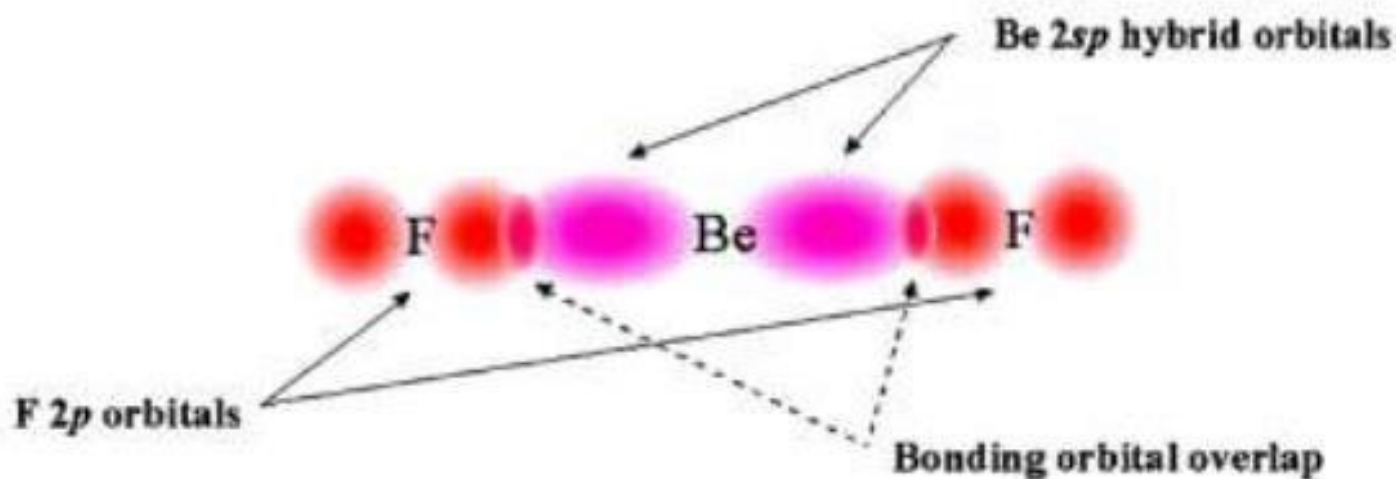
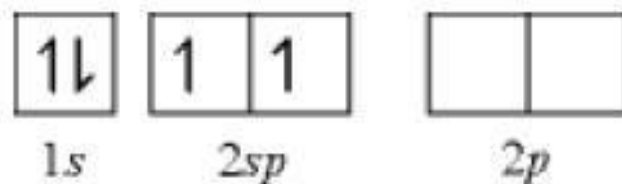
This would actually result in two unpaired electrons, one in a 2s orbital and another in a 2p orbital

The Be atom can now form two covalent bonds with fluorine atoms
We would not expect these bonds to be identical (one is with a 2s electron orbital, the other is with a 2p electron orbital)

We can combine wavefunctions for the 2s and 2p electrons to produce a "hybrid" orbital for both electrons. This hybrid orbital is an **"sp" hybrid orbital**



The orbital diagram for this hybridization would be represented as:



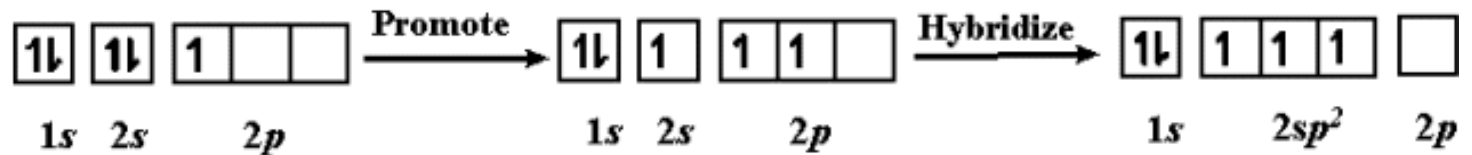


sp^2 and sp^3 Hybrid Orbitals

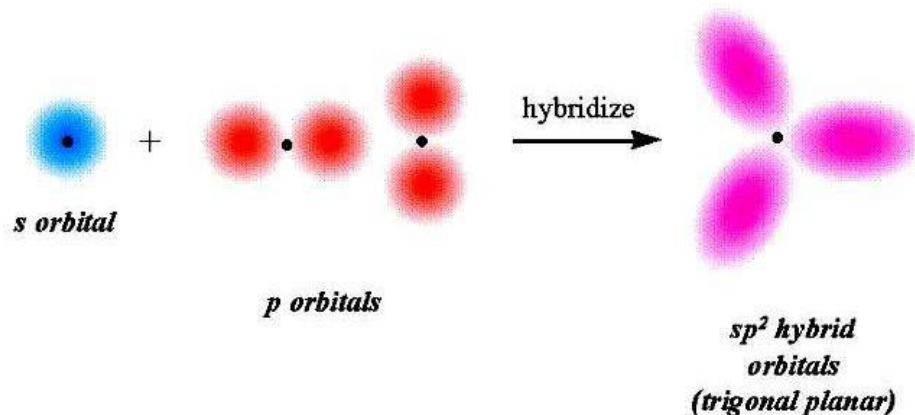
Whenever orbitals are mixed (hybridized):
The number of hybrid orbitals produced is equal to the sum of the orbitals being hybridized. Each hybrid orbital is identical except that they are oriented in different directions

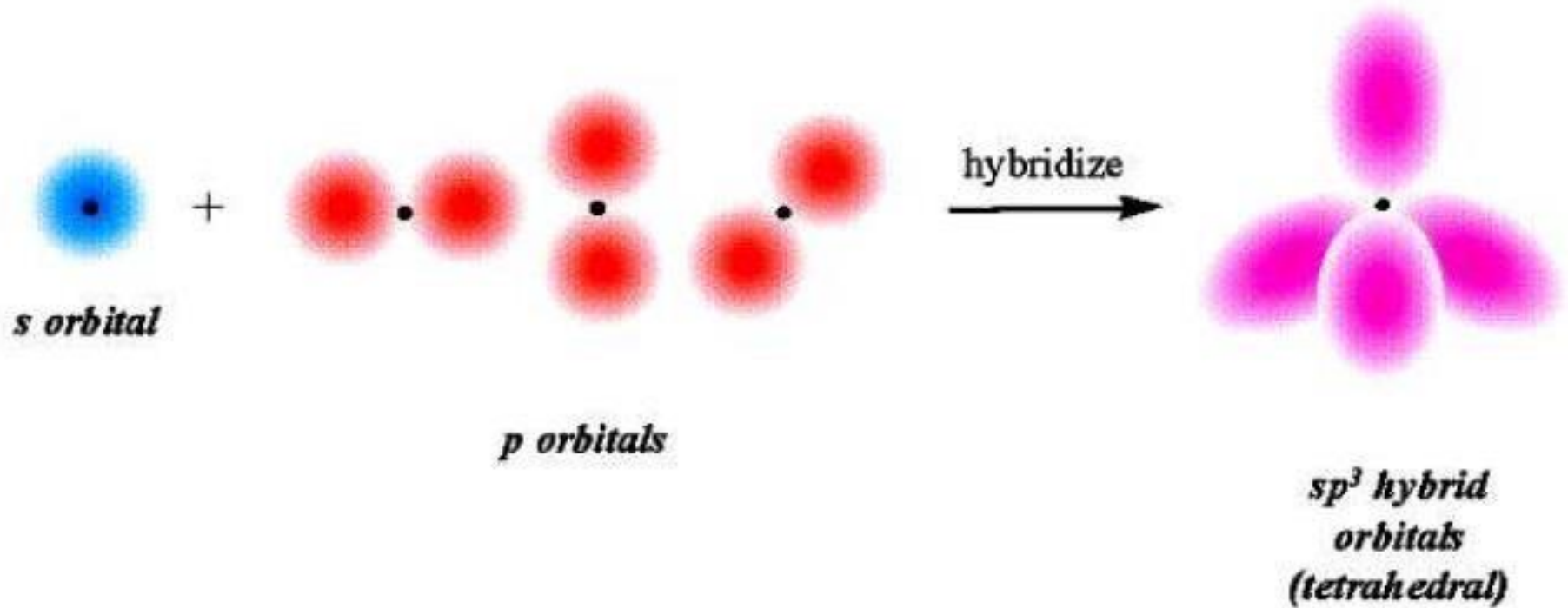
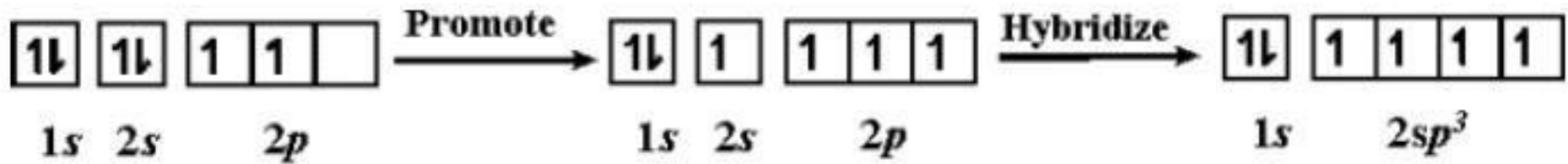


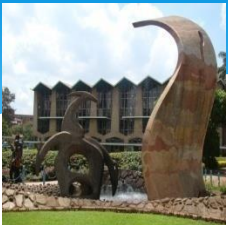
Hybridization of Boron



The three sp^2 hybrid orbitals have a trigonal planar arrangement to minimize electron repulsion





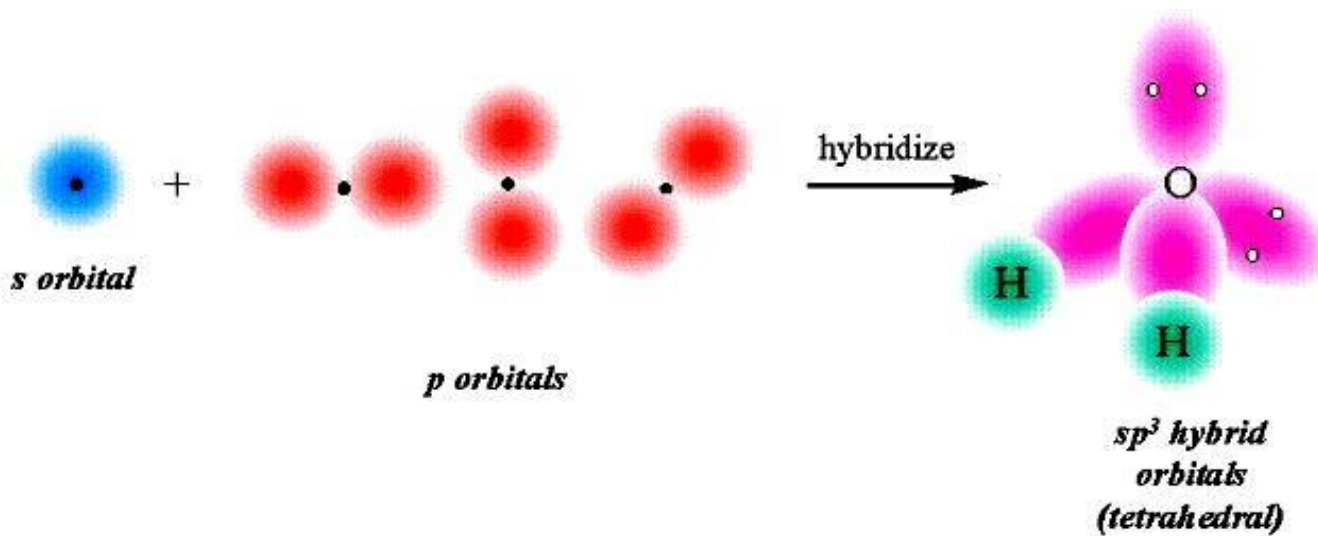
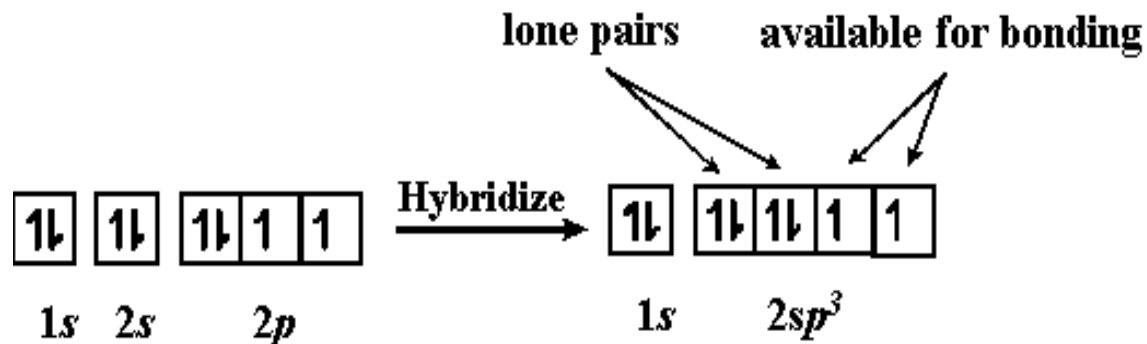


More details



Water (H_2O)

Oxygen =



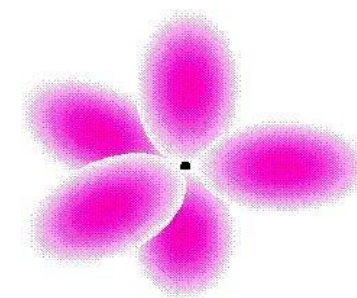
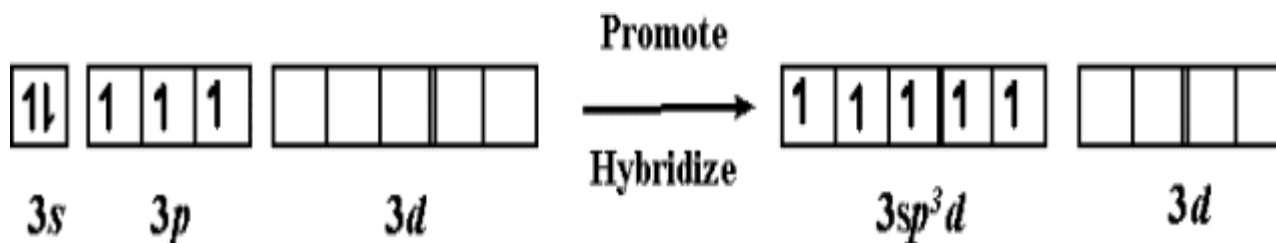


Hybridization Involving d Orbitals



Atoms in the third period and higher can utilize *d* orbitals to form hybrid orbitals

Phosphorous pentafluoride, PF_5



Trigonal Bipyramidal

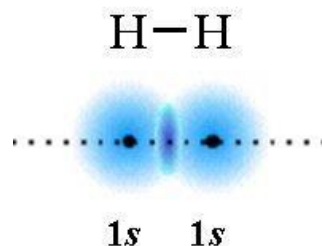
Geometry	Number of Orbitals	Hybrid Orbitals
Linear	2	sp
Trigonal Planar	3	sp^2
Tetrahedral	4	sp^3
Trigonal Bipyramidal	5	sp^3d
Octahedral	6	sp^3d^2



Multiple Bonds and Orbital Overlaps

Sigma (σ) and pi (π) bonds

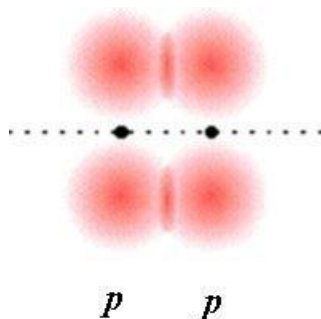
The "inter-nuclear axis" is the imaginary axis that passes through the two nuclei in a bond:



Bonds in which the electron density is symmetrical about the inter-nuclear axis are termed "sigma" or " σ " bonds

In multiple bonds, the bonding orbitals arise from a different type arrangement:

- Multiple bonds involve the overlap between two p orbitals
- These p orbitals are oriented perpendicular to the inter-nuclear (bond) axis



This type of overlap of two p orbitals is called a "pi" or " π " bond. Note that this is a single π bond (which is made up of the overlap of two p orbitals)





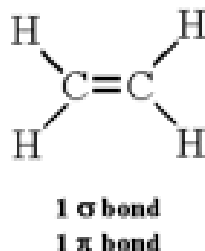
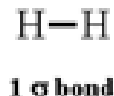
In π bonds:

- The overlapping regions of the bonding orbitals *lie above and below* the inter-nuclear axis (there is no probability of finding the electron in that region)
- The size of the overlap is smaller than a σ bond, and thus the bond strength is typically less than that of a σ bond

A single bond is composed of a σ bond.

A double bond is composed of *one* σ bond and *one* π bond.

A triple bond is composed of *one* σ bond and *two* π bonds.



C_2H_4 (ethylene)

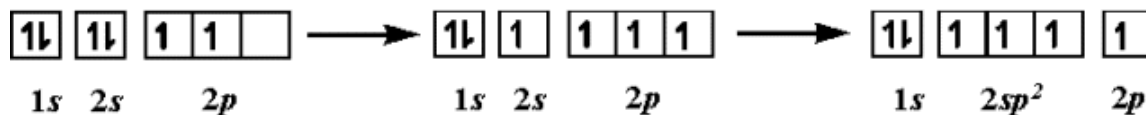
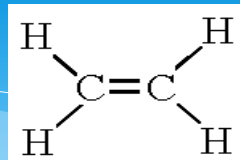
The arrangement of bonds suggests that the geometry of the bonds around each carbon is *trigonal planar*

- Trigonal planar suggests sp^2 hybrid orbitals are being used (these would be σ bonds)

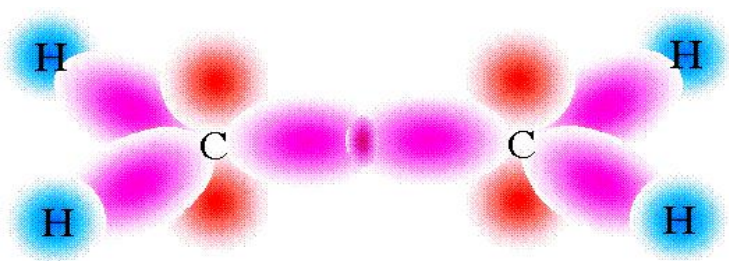
What about the electron configuration?



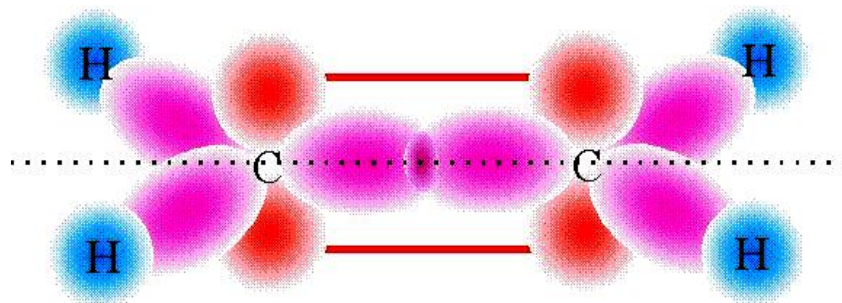
Carbon: $1s^2 2s^2 2p^2$



Thus, we have an extra unpaired electron in a p orbital available for bonding
 This extra p electron orbital is oriented perpendicular to the plane of the three sp^2 orbitals (to minimize repulsion):



The unpaired electrons in the p orbitals can overlap one another above and below the internuclear axis to form a covalent bond



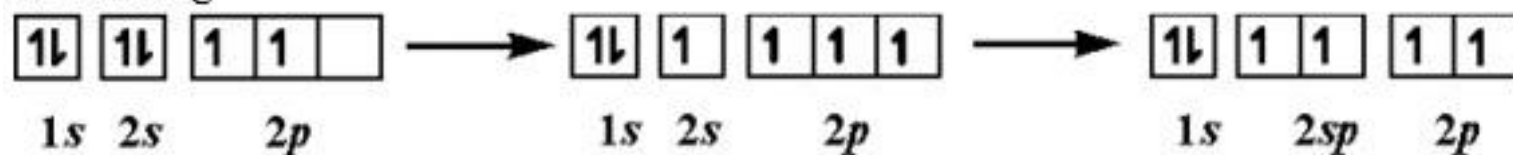
This interaction above and below the inter-nuclear axis represents the single π bond between the two p orbitals



C_2H_2 (acetylene)

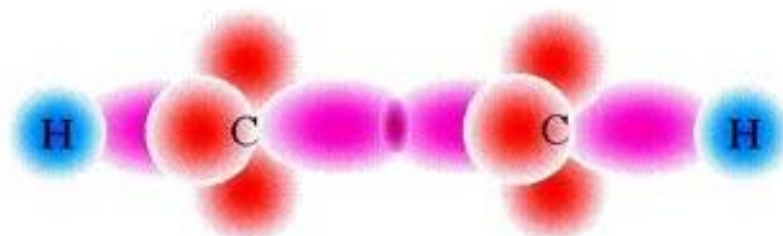


The linear bond arrangement suggests that the carbon atoms are utilizing sp hybrid orbitals for bonding

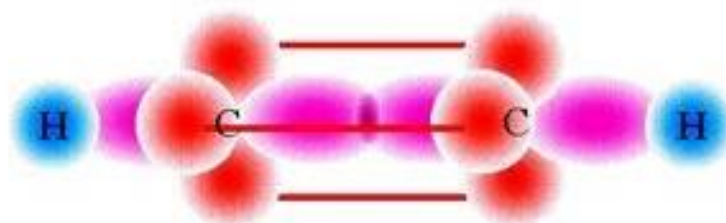


This leaves two unpaired electrons in p orbitals

To minimize electron repulsion, these p orbitals are at right angles to each other, and to the inter-nuclear axis:



These p orbitals can overlap to form two π bonds in addition to the single σ bond (forming a triple bond)





The Molecular-Orbital Theory

The molecular-orbital approach starts by considering the **stable molecular system in which the nuclei are in their equilibrium positions, and produces molecular wavefunctions that describe the molecular energy states (molecular orbitals) into which electrons are placed.**



From quantum mechanics,

1. The molecular Schrödinger equation is 'solved' to provide acceptable molecular wavefunctions that describe molecular energy states (molecular orbitals).
2. These molecular orbitals will embrace both nuclei. (In the case of polynuclear molecules the molecular orbitals will in principle involve all the nuclei.).
3. The molecular orbitals will be associated with molecular quantum numbers.
4. The molecular wavefunctions will have the same significance as do atomic wavefunctions, i.e. $\psi^2 d\tau$ is proportional to the probability of finding the electron in a given volume element of space $d\tau$.
5. Each molecular wavefunction corresponds to a definite energy value, and the sum of the energies of individual electrons in the molecular orbitals, after correction for interaction, represents the total energy of the molecule.
6. The 'Aufbau' principle may be applied. The electrons are filled into the available molecular orbitals one at a time, the molecular orbital of lowest energy being filled first.
7. Each electron has a spin, and by the Pauli Exclusion Principle, each molecular orbital can accommodate a maximum of two electrons, provided their spins are opposed.



THE LCAO METHOD (The Linear Combination of Atomic Orbitals)



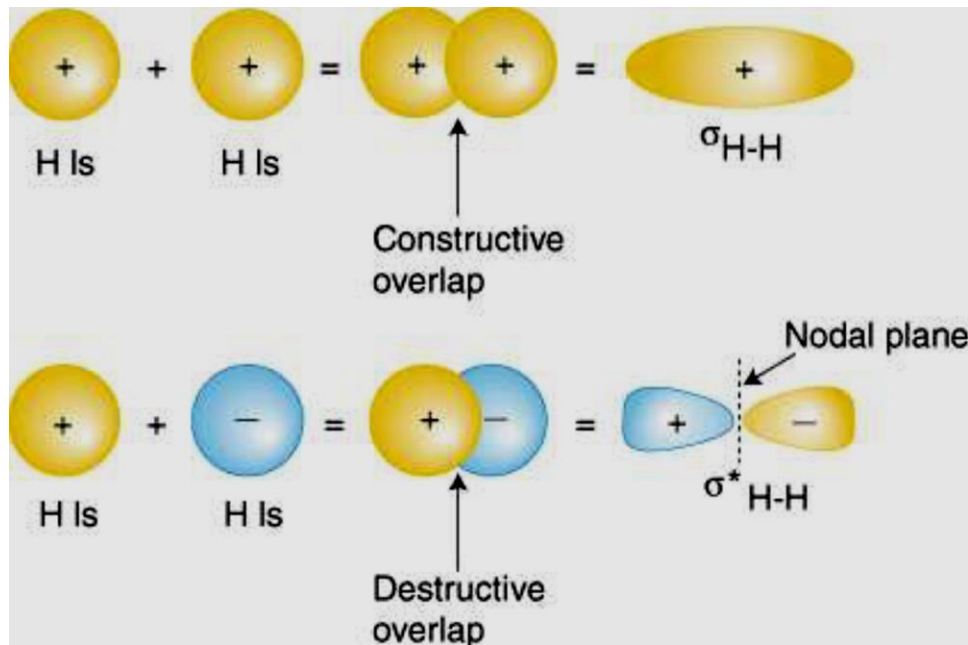
Construction of molecular orbitals may be done by adding atomic orbitals as follows:

1. All the atomic orbitals in the valence shells of the interacting atoms are combined in pairs to give molecular orbitals for the interacting nuclei.
2. Each pair consists of one atomic orbital from each of the interacting atoms. The atomic orbitals may or may not be originally occupied.
3. For combination of the atomic orbitals to occur the following conditions have to be met:

- a) The combining orbitals must be associated with approximately the same energy
- b) The combining orbitals must be capable of overlapping
- c) The combining orbitals must be of the same symmetry



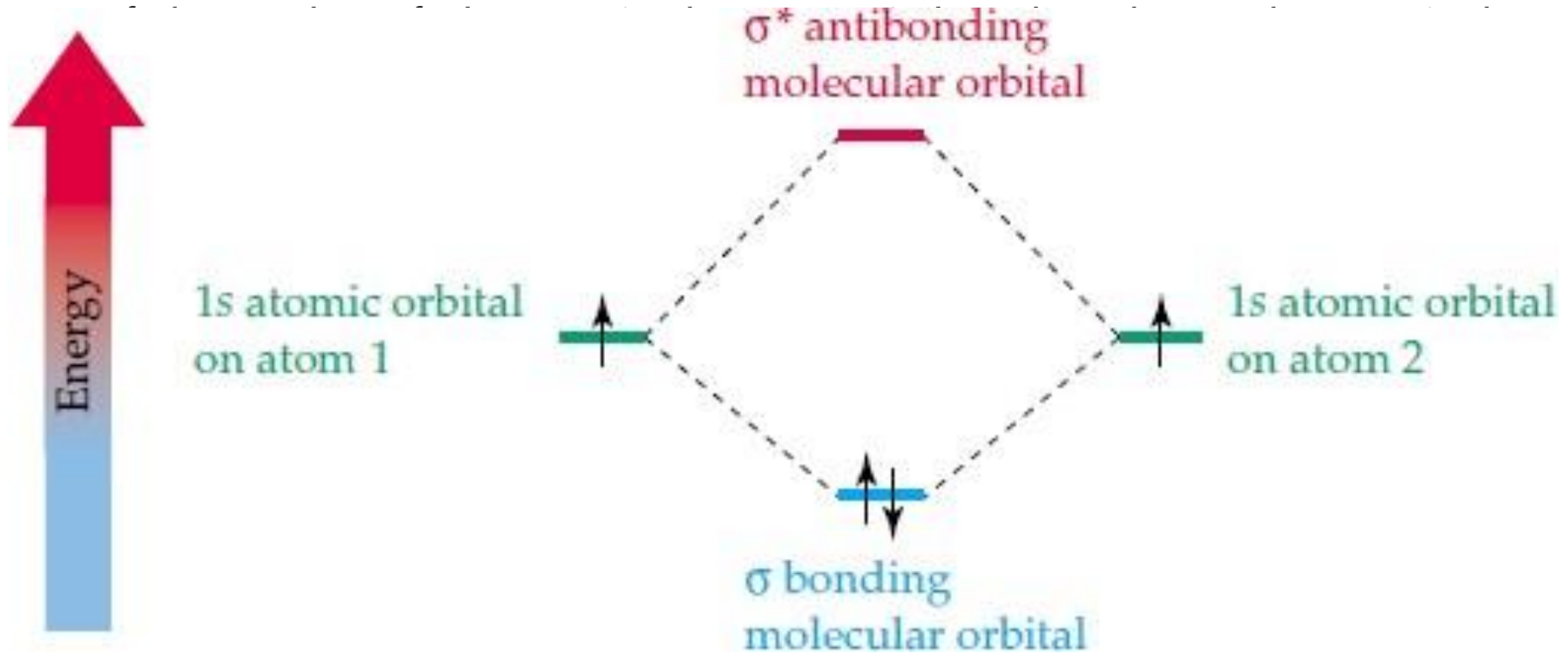
4. The number of molecular orbitals produced is equal to the number of combining atomic orbitals. Half will be bonding and half anti-bonding. (Constructive interference of the wave-functions leads to bonding molecular orbitals while destructive interference leads to anti-bonding molecular orbitals).



Two 1s orbitals combine to form a bonding and an antibonding M.O.



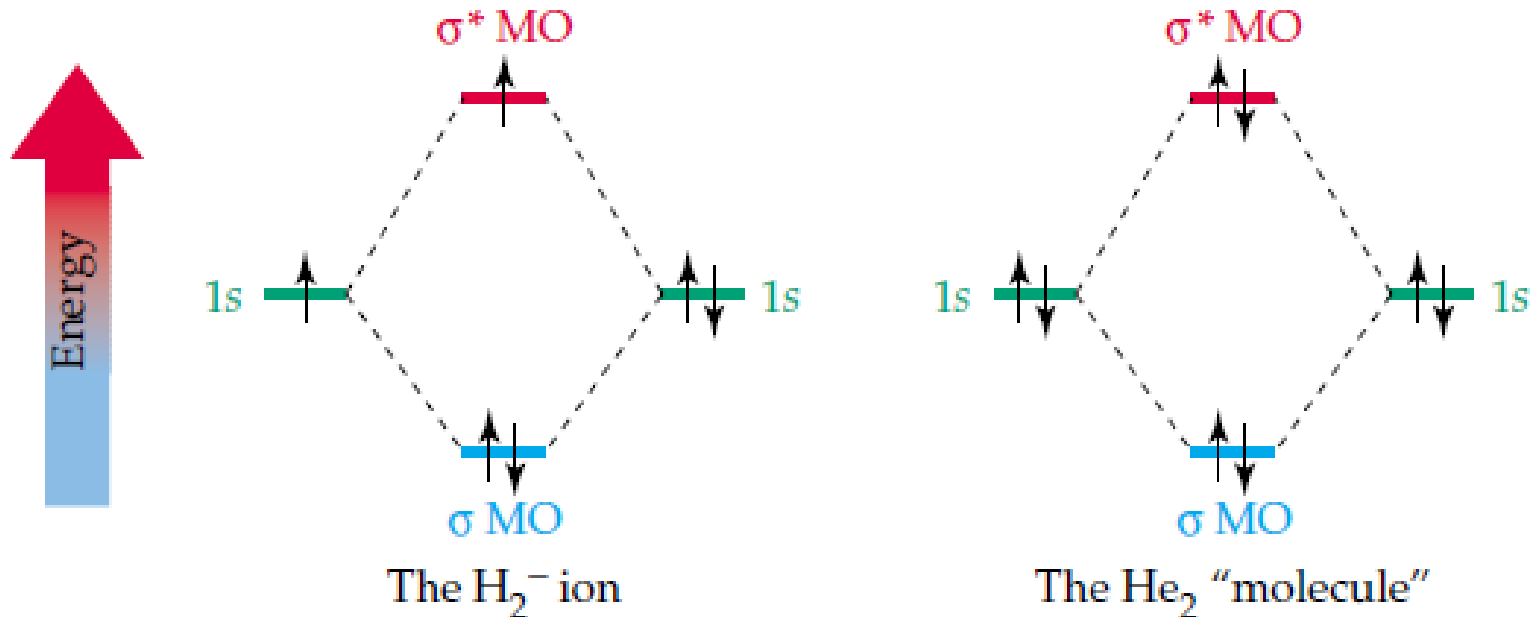
5. The molecular orbitals are arranged in ascending order of energy (which can be represented by energy level diagrams) and all the electrons originally occupying the valence shells are redistributed into these orbitals according to the **Aufbau** principle and obeying the **Pauli exclusion** principle.



Orbital correlation diagram for hydrogen



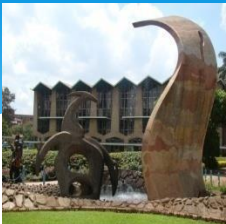
An orbital correlation diagram for a hypothetical He-He molecule



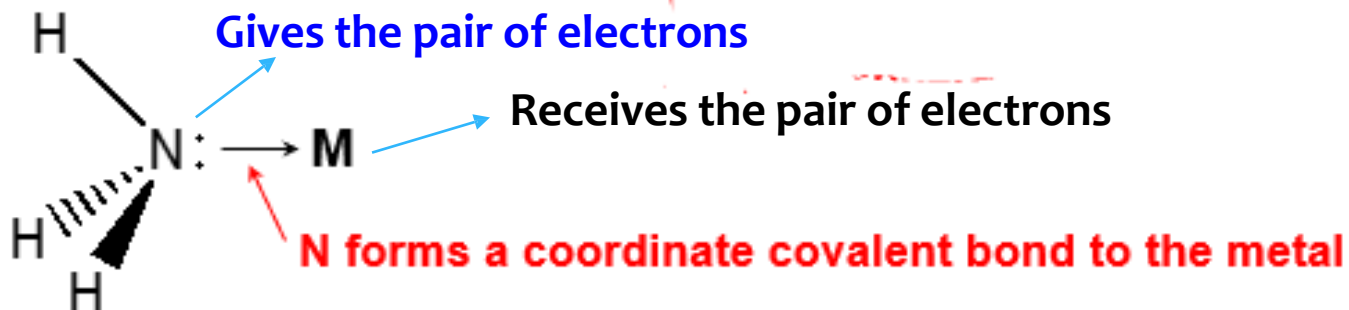
Bond Order

In any molecule bond order is given by:

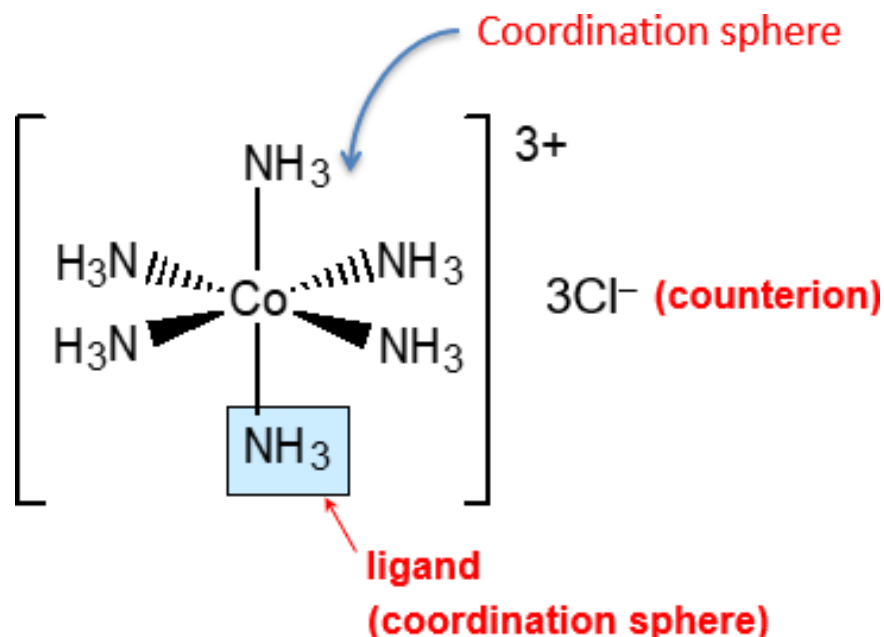
$$\text{Bond order} = \frac{\sum \text{Electrons in bonding MO} - \sum \text{Electrons in antibonding MO}}{2}$$



Coordinate bonding



This kind of bonding leads to coordination compounds especially between metals and inorganic ions or molecules



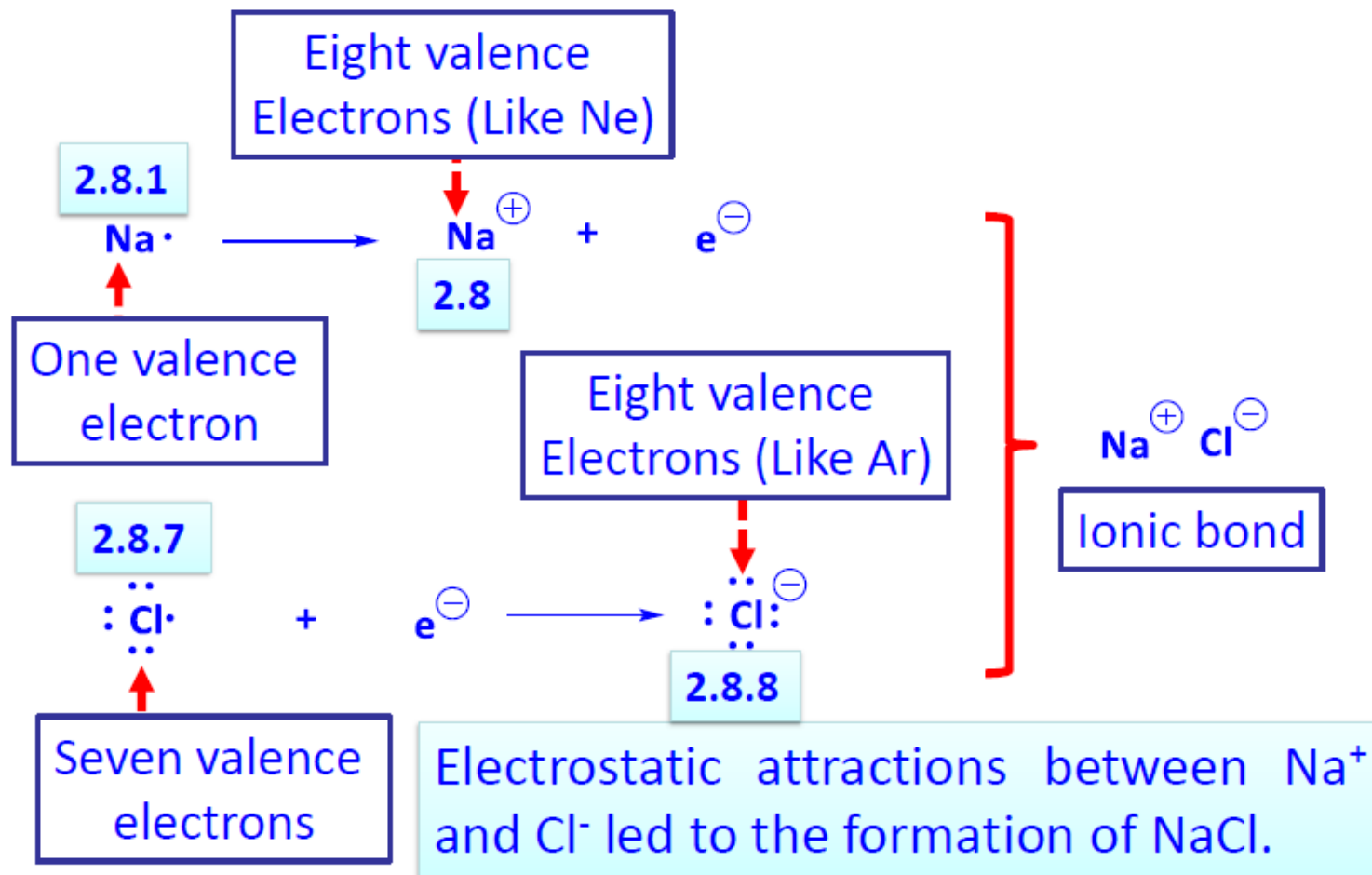


Ionic bonding



Ionic bonds result from the transfer of electrons from one element to another.

Sodium chloride, NaCl, is an example of an ionic compound.





Ionic compounds



Many compounds can be thought of as a collection of ions (M^{n+} , X^{n-}) held together electrostatically. This idea arose out of experiments by Arrhenius looking at the conductivity of solutions prepared by dissolving ionic compounds in water.

Characteristics of ionic compounds

- Most simple ionic compounds tend to form hard and brittle crystals
- They usually have high melting points - several hundred or thousand Kelvin.
- However, salts that are liquid at room temperature have been prepared using organic cations.
- When molten they conduct electricity
- Most dissolve in high polarity solvents to form conducting solutions

Fajan's rules

- Small highly charged cations are more polarizing
- Large highly charged anions are more polarizable
- Polarization is favoured for cations that do not have a noble gas electron configuration e.g. Ag^+ , Zn^{2+} , Cd^{2+} , Hg^{2+} etc.



Lattice energy

Energy evolved when one gram mole of a crystal is formed from gaseous ions or
Energy required to convert one gram mole of a crystal into its constituent gaseous ions.
Lattice energies cannot be measured directly but experimental values are obtained from thermodynamic data using the Born-Haber cycle. Theoretical values can be calculated using the Born-Lande equation:

$$U = -\frac{N_o A z^+ z^- e^2}{4\pi\epsilon_o r_o} \left(1 - \frac{1}{n}\right)$$

ϵ_o = permittivity of free space = $8.854 \times 10^{-12} \text{ Fm}^{-1}$,

A = the Madelung constant which depends on the geometry of the crystal

N_o = the Avogadro constant = $6.023 \times 10^{23} \text{ mol}^{-1}$

Z^+ , Z^- are the charges on the cations and anions, respectively.

r_o = the equilibrium inter-ionic distance

e = the charge on an electron

n = the Born exponent

This equation allows the lattice energy of any crystal to be calculated from a knowledge of the geometry of the crystal, and hence the Madelung constant, the charges z^+ and z^- and the inter-ionic distance.



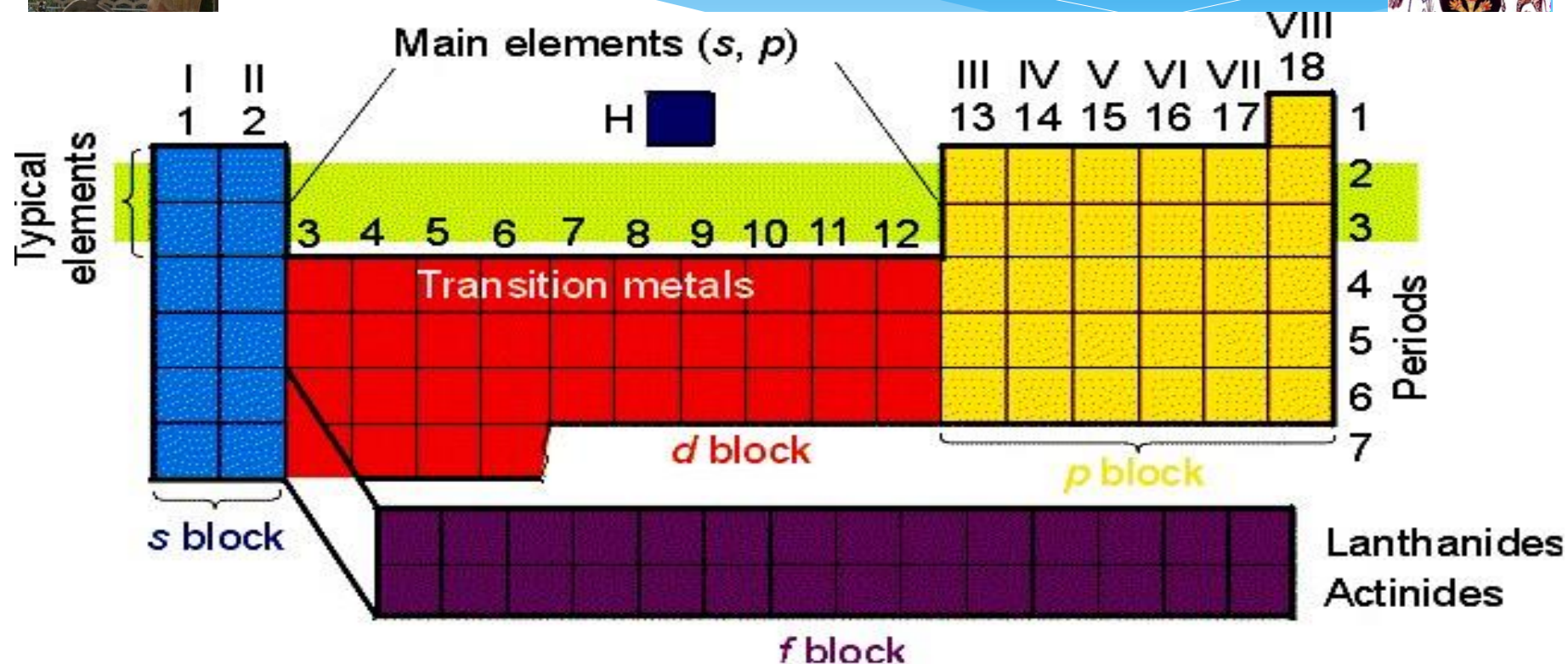
Several important points arise from the Born-Lande equation:



- ✓ The inter-ion distance is the denominator of the equation. Thus the smaller the value of r_0 the greater the amount of energy evolved when the crystal lattice is formed, and hence the more stable the crystal will be. U is proportional to $1/r$. When the inter-ionic distance becomes small enough for the ions to touch, they begin to rebel each other due to the mutual repulsion of the electron clouds on the two ions.
- ✓ The lattice energy depends on the product of the ionic charges, and U is proportional to $(Z^+.Z^-)$.
- ✓ The experimental and theoretical values for the alkali metal halides and the oxide and halides of alkaline earth metals (except Be), all agree to within 3%. However this close agreement does not in itself prove that the equation itself or the assumptions on which it's based, are correct. There are two opposing factors in the equation. Increasing the inter-ionic distance r reduces the lattice energy. It is almost impossible to change the r without changing the structure and hence the Madelung constant A . Increasing A increases the lattice energy: hence the effects of changing r and A may largely cancel each other.
- ✓ Crystals with a high lattice energy usually melt at high temperatures and are very hard. High lattice energy is favoured by a small inter-ionic distance and a high charge on the ions



Selected Periodic Trends



The structure of the periodic table

Question

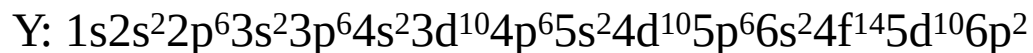
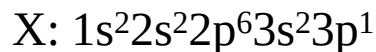
Elements X and Y (not their actual symbols) have atomic number 13 and 82 respectively. To which period and groups do they belong?



Solution



The electronic configuration of:



Period numbers

X the largest principal quantum number, n , is 3

Implying period 3

Y the largest principal quantum number, n , is 6

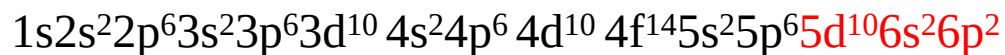
Implying period 6

Group numbers

X: Electrons already arranged in ascending order according to the value of n .

Since it belongs to period 3. Group number $= 10 + 3 = 13$

Y: Arranging electrons according to n



The last p^6 to be filled is $5p^6$. The number of electrons after that is $= 10 + 2 + 2 = 14$



Atomic properties such as effective nuclear charge, atomic radii, ionization energies, electron affinity and electronegativity are important in accounting for the chemical properties of an element.



Shielding

In atoms with more than one electron, the effect of electron-electron repulsion depends on where the various electrons are located in the atom.

Electrons in the outer shell (higher n) are pushed away by electrons in the inner shell (lower n). As a result, the net nuclear charge (or effective nuclear charge, Z_{eff}) felt by an outer electron is substantially lower than the actual nuclear charge (Z). We say that the outer electrons are **shielded** from the full charge of the nucleus by the inner electrons.

Electrons in the same shell (same n) have an immediate effect in shielding the nuclear charge.

Electrons in the outer shell (higher n) do not shield the nuclear charge from the inner electrons (lower n)



In 1930, Slater formulated a set of rules for the **effective nuclear charge** felt by electrons in different atomic orbitals based on experimental data. The effective nuclear charge Z_{eff} can be calculated by the following equation:

$$Z_{\text{eff}} = Z_{\text{actual}} - S$$

Z: atomic number (number of protons): Slater screening constant;

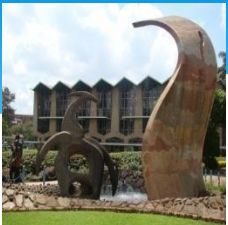
The values of S are estimated as follows:

Write out the electron configuration of an element in groups of: (1s), (2s, 2p), (3s, 3p), (3d), (4s, 4p), (4d), (4f), (5s, 5p) etc.

Electrons with higher n do not shield those in lower n ; $S = 0$.

If the electron in question resides in an s or p orbitals:

- Each of the other electron in the same ns , np group, $S = 0.35$.
- Each electron in the $(n - 1)$ shell, $S = 0.85$.
- Each electron in the $(n - 2)$ and lower shell, $S = 1.00$
- **If the electron in question resides in a d or f orbitals:**
 - Each of the other electron in the same nd , nf group, $S = 0.35$.
 - Each electron in the $(n - 1)$ shell, $S = 1.00$



Sizes of Atoms and Ions



Since electrons can only be located by probability, there is no real boundary to an atom. Radii of atoms and ions are obtained experimentally from measuring bond lengths in molecules, metals and ionic crystals by X-ray diffraction.

Covalent radii r_{cov} : defined as the half-distance between the nuclei of two atoms of the same element joined by a covalent bond. E.g., Cl_2 : bond distance = 198 pm, atomic radius of Cl = $(198)/2 = 99$ pm.

Van der waal radius r_{vdw}

Defined as the half-distance between the nuclei of two non-bonded atoms or molecules. Obtained from the closest approach between two non-bonded atoms or molecules in the solid state. E.g., van der waal radius of Cl = 180 pm.

In general, $r_{\text{vdw}} > r_{\text{cov}}$

Ionic radius:

Defined as the distance between the nuclei of adjacent cations and anions in a purely ionic lattice.

In general, $r_{\text{anions}} > r_{\text{cov}} > r_{\text{cation}}$ for any given element.



Electronegativity χ



The tendency of an atom to attract electrons to itself in a chemical bond is referred to as **electronegativity**. The greater the electronegativity of an atom, the more strongly the atoms attracts the electrons of a bond.

Electronegativity generally decreases from top to bottom in a group; because with each successive shell, the bonding electrons are further from the nucleus.

Electronegativity χ of some elements

Group Period	1	2	13	14	15	16	17	18
1	H 2.2							He -
2	Li 1.0	Be 1.5	B 2.0	C 2.5	N 3.1	O 3.5	F 4.1	Ne -
3	Na 1.0	Mg 1.2	Al 1.3	Si 1.7	P 2.1	S 2.4	Cl 2.8	Ar -
4	K 0.9	Ca 1.0	Ga 1.8	Ge 2.0	As 2.2	Se 2.5	Br 2.7	Kr 3.1
5	Rb 0.9	Sr 1.0	In 1.5	Sn 1.7	Sb 1.8	Te 2.0	I 2.2	Xe 2.4

Increase of electronegativity $\longrightarrow$



Using χ to estimate bond polarity

Atoms with similar electronegativity ($\Delta\chi \leq 0.4$) form non-polar bonds.

Atoms whose electronegativity differs by more than 2 form ionic bonds.

Atoms whose electronegativity differs by less than 2 form covalent bonds.